

SYSTEMS ANALYSIS FUNDAMENTALS AND ORGANIZATIONAL CONTEXT



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MODULE 1

LEARNING OBJECTIVES

What you will learn

ILO 1: Explain systems concepts, analyst roles, ethical responsibilities, and predictive, agile, prototyping, cloud, and hybrid approaches.

ILO 2: Analyze organizational context, stakeholders, problem or opportunity evidence, root causes, and expected value.

Before you begin

[AI-PROHIBITED] Complete the Diagnostic Assessment for Summative Assessment 1 in Canvas before opening Module 1. It is ungraded and used only for readiness mapping and targeted support.



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Question of understanding

Why do useful systems still fail after deployment—even when the software works as coded?



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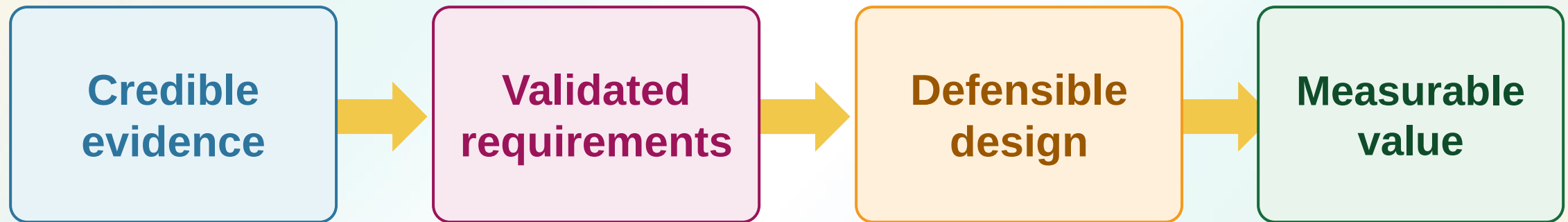
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SYSTEMS, ROLES, AND DEVELOPMENT APPROACHES

Disciplined analysis connects evidence to value

Good systems begin with evidence, not with a favorite technology or feature list.



Planning reduces rework, unmanaged risk, stakeholder conflict, and solutions that address symptoms instead of causes.

Analysis and design answer different questions

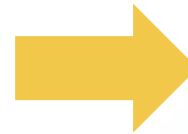
Both are necessary, but they solve different parts of the decision.

SYSTEMS ANALYSIS

What problem or opportunity exists?

Who is affected?

What must the system accomplish?



SYSTEM DESIGN

How will the validated solution work?

How will data, interfaces, controls, and components fit together?



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Disciplined SAD improves decision quality

The goal is not to standardize one process; it is to make choices traceable and fit for context.

**Understand
context**

**Separate causes
from symptoms**

Align stakeholders

**Reduce risk and
rework**

**Produce traceable
decisions**

Each outcome depends on evidence, stakeholder participation, and documented judgment.



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Compare and tailor development approaches

No method is automatically best; fit depends on uncertainty, risk, feedback, and delivery conditions.

Predictive

Agile

Prototyping

Cloud
lifecycle

Hybrid

TAILOR USING: requirement stability • stakeholder access • compliance • technical risk • release cadence • team capability



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Problem framing turns evidence into direction

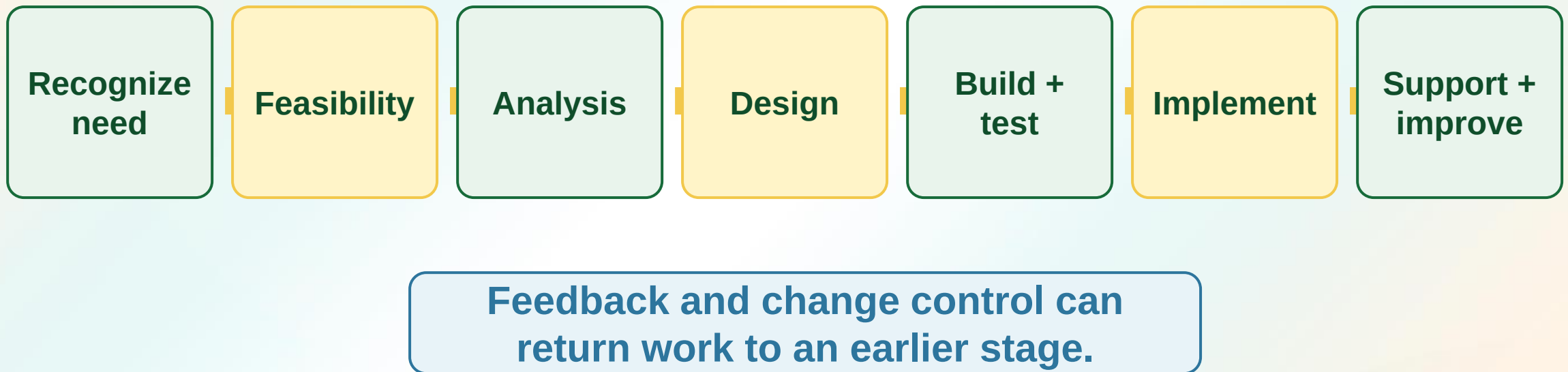
Move from what is happening to why it matters before proposing a solution.



A title becomes defensible only when this chain is supported by organizational evidence.

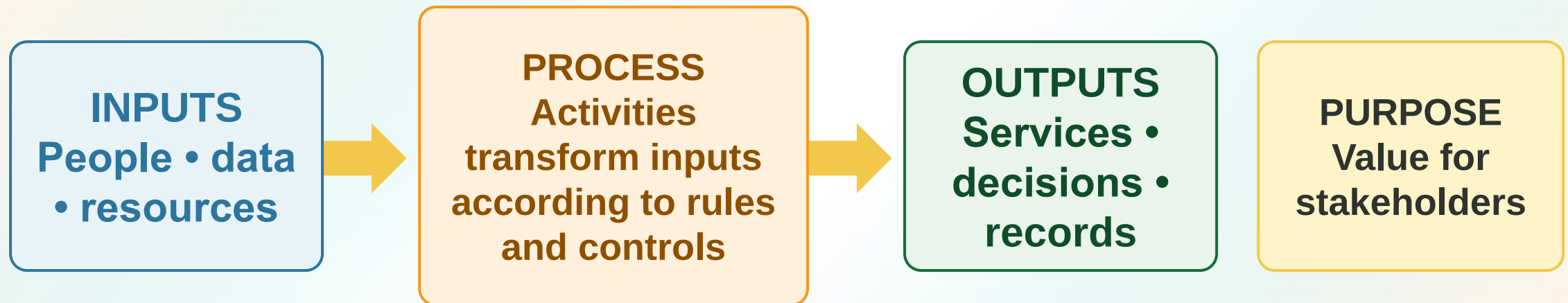
One predictive SDLC example

Use this sequence when scope and requirements can be established with reasonable stability; feedback can still trigger revision.



A system is interrelated parts working toward a purpose

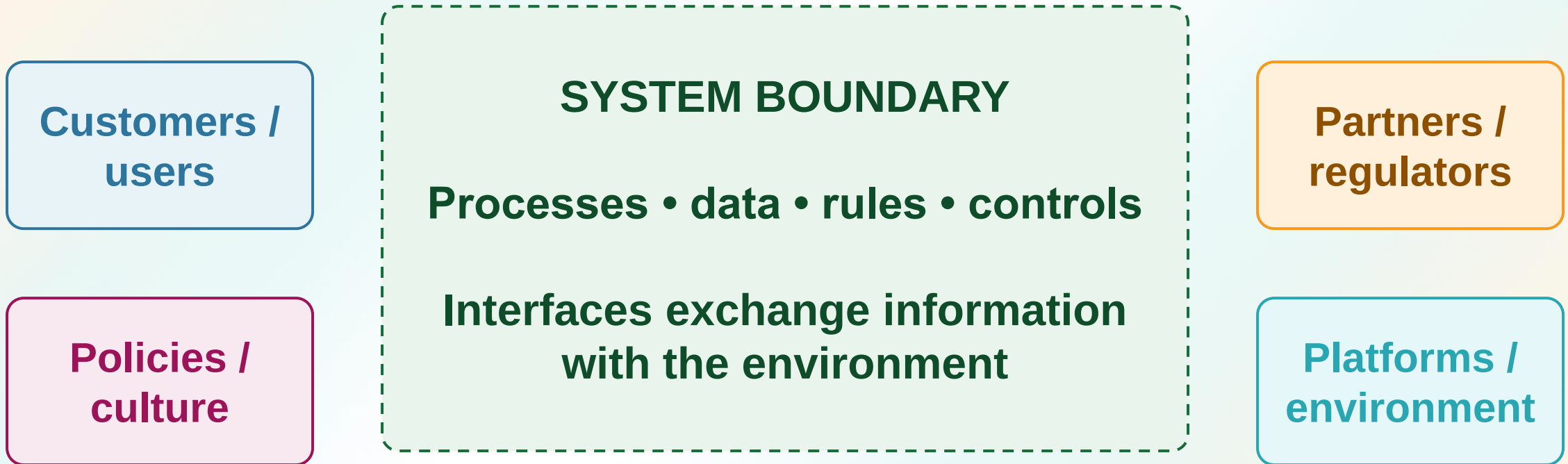
A useful model shows what enters, what changes, what leaves, and how performance is corrected.



FEEDBACK + CONTROL compare actual results with expectations and guide correction.

Boundaries clarify control and interaction

Define the system, external actors, interfaces, feedback, and assumptions before setting scope.



Analysts connect people, evidence, and decisions

The role changes with the engagement, but the analyst remains accountable for traceable reasoning.

Consultant

Clarifies a need
and
recommends a
direction

Supporting expert

Contributes
specialized
analysis or
methods

Change agent

Helps
stakeholders
adopt
improved work

Facilitator

Builds shared
understanding
and resolves
conflict



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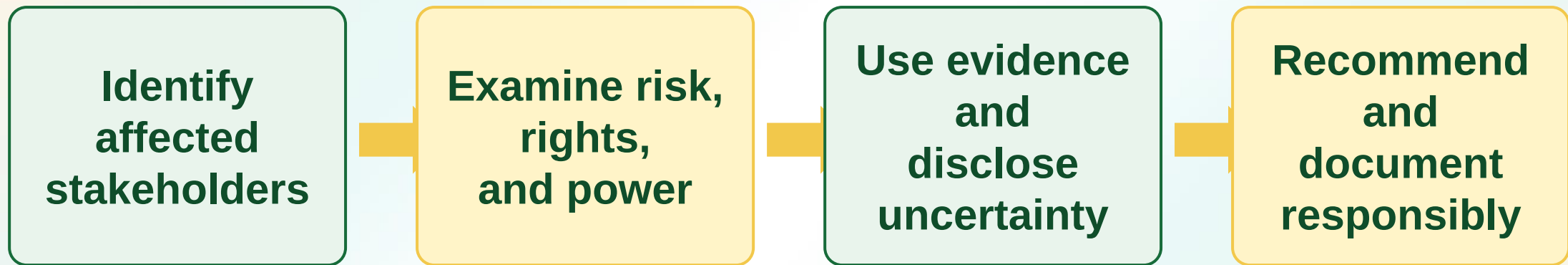
Effective analysts combine five competencies

Technical knowledge alone is insufficient when the real challenge involves people, policy, and competing interests.



Ethical analysis protects stakeholders

Professional judgment asks what is permitted, what is responsible, and whose interests may be overlooked.



RED FLAGS: hidden conflicts • unnecessary data collection
• discriminatory impact • unsupported claims • concealed AI assistance

Responsible design begins with four lenses

Security, privacy, accessibility, and responsible AI use shape requirements—not just final testing.

SECURE BY DESIGN
Threats and controls

PRIVACY BY DESIGN
Purpose and
minimization

**TRUSTWORTHY
SOLUTION**

ACCESSIBILITY
Inclusive interaction

RESPONSIBLE AI
Disclosure and
validation



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DEVELOPMENT APPROACHES

Five approach families fit different conditions

Compare the logic of each approach before selecting or combining them.

Predictive

Stable scope
and formal
gates

Agile

Frequent
change
and
feedback

Prototyping

Unclear
needs
or
interaction

Cloud lifecycle

Continuous
delivery
and
operations

Hybrid

Mixed
certainty
and
constraints



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Predictive development follows planned progression

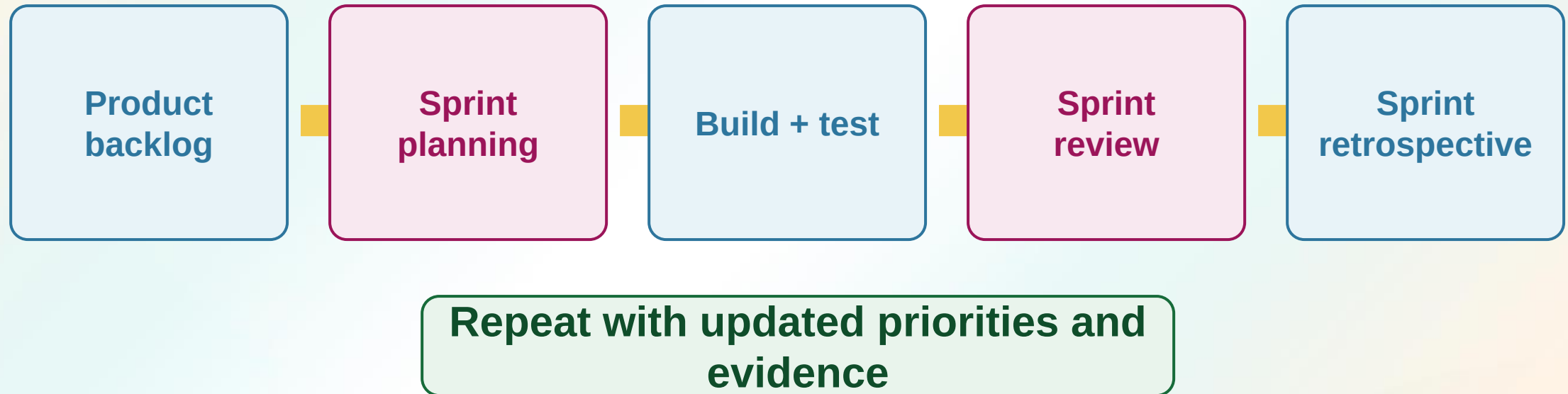
Use it when requirements are comparatively stable and formal documentation or approval gates matter.



Strength: traceability and predictability • Risk: late feedback can make change expensive

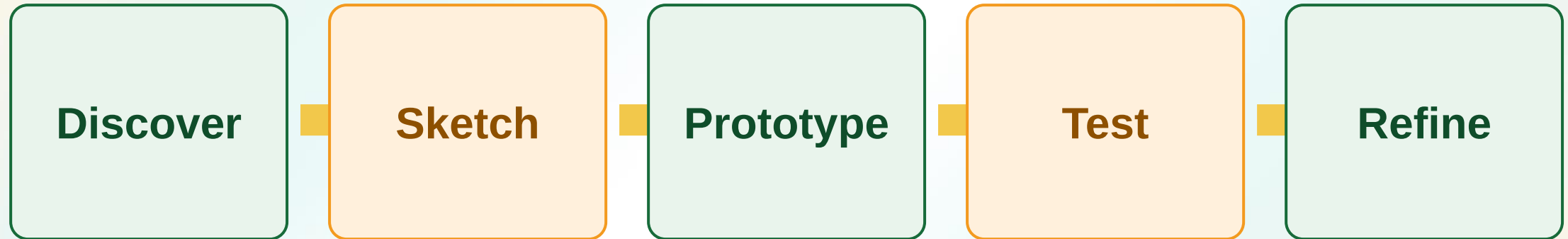
Agile and Scrum organize short learning cycles

A product backlog is refined through time-boxed work, review, and retrospective improvement.



Prototyping reduces uncertainty through interaction

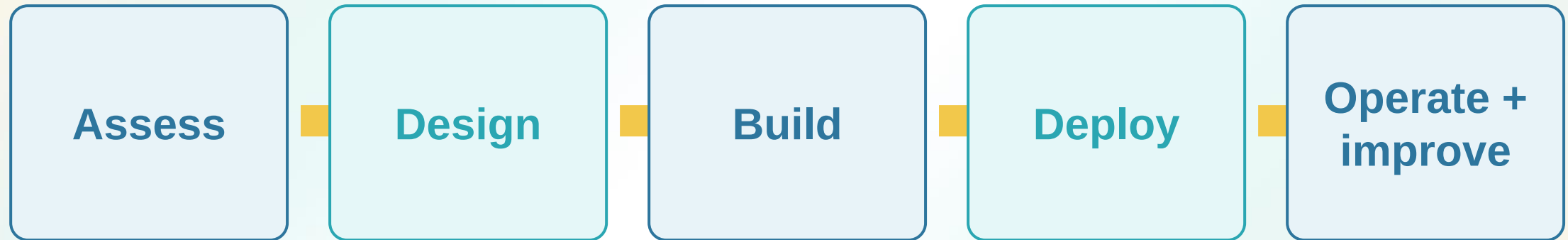
A prototype is a learning instrument; it does not automatically become the production system.



Validate assumptions about workflow, information, usability, and value before committing to detailed design.

Cloud development links delivery and operations

Architecture, deployment, observability, security, and cost decisions evolve together.



**Continuous concerns: security • privacy •
resilience • observability • cost**

Hybrid delivery combines methods with a rationale

Combine methods only when the boundaries, handoffs, governance, and risks are clearly defined.



A hybrid is not “use everything.” Each part must solve a specific delivery condition.

Select an approach with evidence-based criteria

Use the same case evidence to compare alternatives; document both the choice and the rejected options.

**Scope
stability**

**Stakeholder
access**

**Risk +
compliance**

**Release
cadence**

**Team +
operations**

**Decision record = selected approach + tailoring
rules + evidence + limitations + risks**



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Workshop: compare development approaches

Case condition: Describe the specialization setting, uncertainty, risk, and stakeholder access.

Compare: Predictive, agile, prototyping, cloud, and hybrid approaches using the same criteria.

Decide: Select and tailor one direction; record why the alternatives were rejected.

Defend: Connect the recommendation to credible course and organizational evidence.



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Technical Activity 1: Audit AI advice

Use AI to generate a comparison, then verify, critique, correct, and disclose your responsible use.



[AI-INTEGRATED] Submit prompts, outputs, validation, corrections, and AI Use Disclosure.

ORGANIZATIONAL SYSTEMS AND PROBLEM FRAMING



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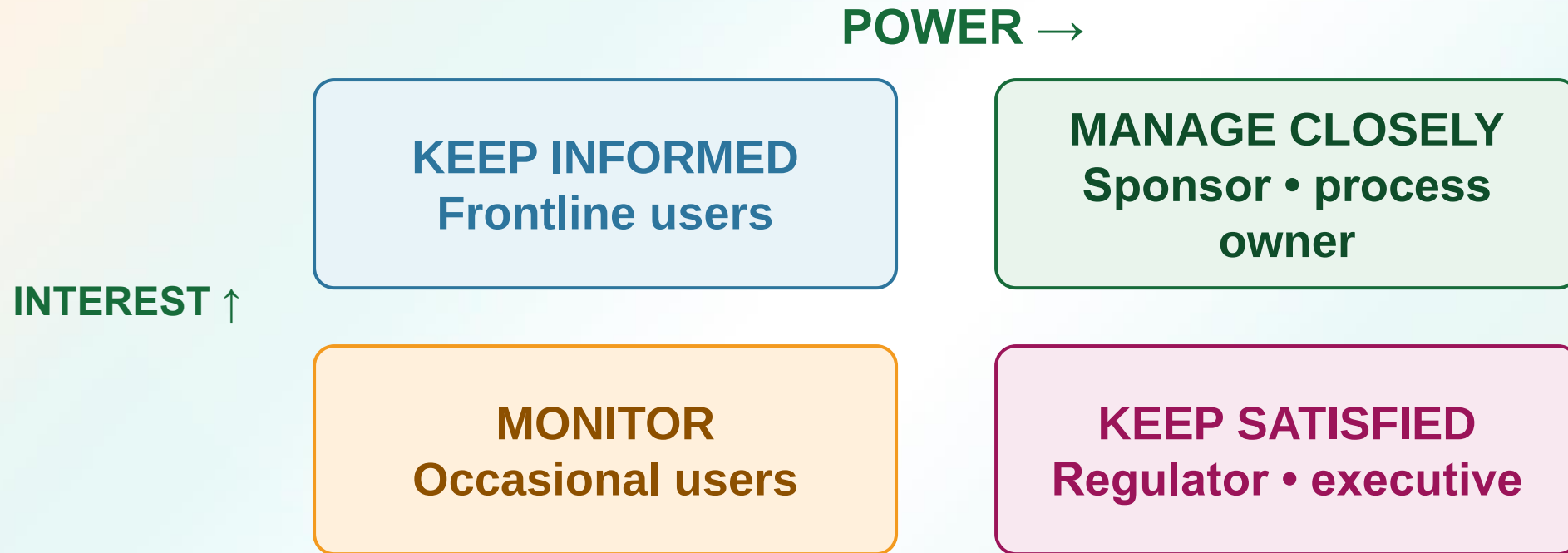
Organizations behave as interacting systems

A change in one element can improve—or destabilize—other parts of the organization.



Stakeholder maps reveal influence and impact

Place stakeholders by power and interest, then plan how each group should participate.



Context diagrams define external relationships

Keep the proposed system as one process at this level; show actors and exchanged information.



Rich pictures capture the messy reality

Show people, activities, information, tensions, risks, and unanswered questions in one shared view.

PEOPLE

Who acts or decides?

PROCESS

What actually happens?

INFORMATION

What is created or lost?

TENSIONS

Where do goals conflict?

Add symbols for delays, workarounds, security concerns, complaints, uncertainty, and opportunities.



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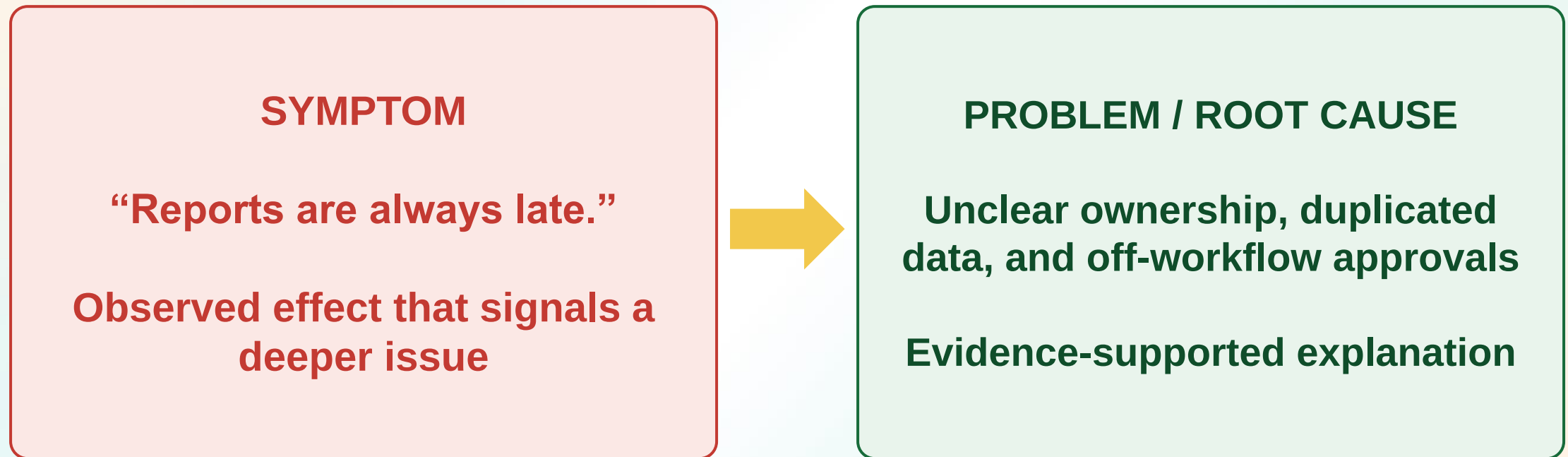


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Symptoms are visible; problems explain persistence

Do not convert the first complaint into a system feature until the underlying cause is supported.



Triangulate evidence before making a claim

A defensible problem statement uses multiple sources and explains contradictions rather than hiding them.

Interviews

Observation

Documents
+
policies

Logs +
records

Surveys +
metrics

TRIANGULATED FINDING
What do the sources consistently support?



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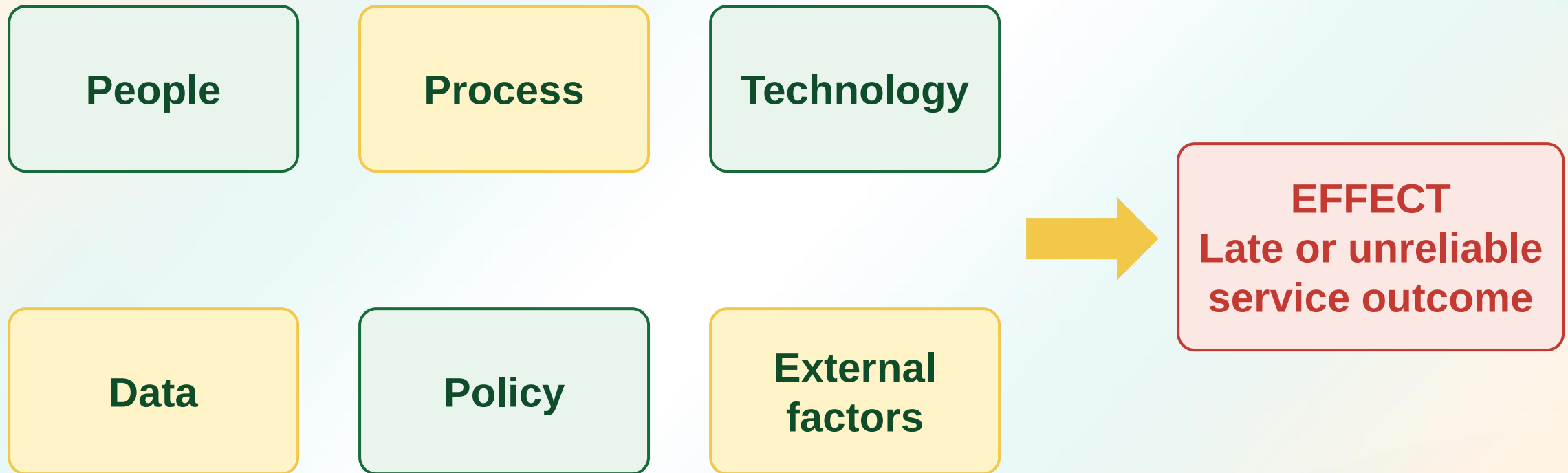


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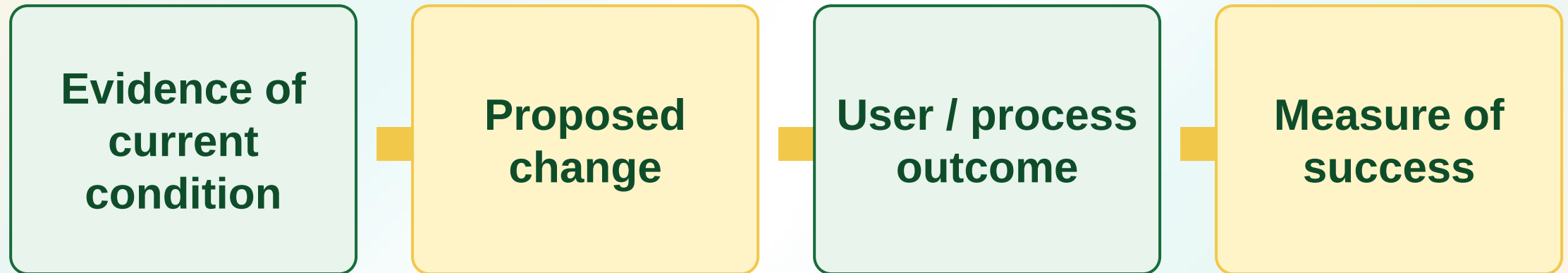
Ishikawa analysis organizes causes for verification

Treat branches as hypotheses; confirm them with evidence before calling them root causes.



Expected value links change to measurable outcomes

State who benefits, what improves, and how success could later be observed.



Example: reduce duplicate encoding and improve the timeliness and traceability of approved records.

SPECIALIZATION STUDIO

Cybersecurity lens: protect the operational workflow

Begin with the transaction and evidence; then identify trust boundaries, abuse paths, and required safeguards.

WORKFLOW
Incident intake
and validation

RISK
False reports,
tampering,
unauthorized
access

**CONTROL
NEED**
Authentication,
integrity,
traceability

VALUE
Faster,
defensible
response
decisions

Project framing: operational workflow + verified threat + justified control + measurable security/mission outcome



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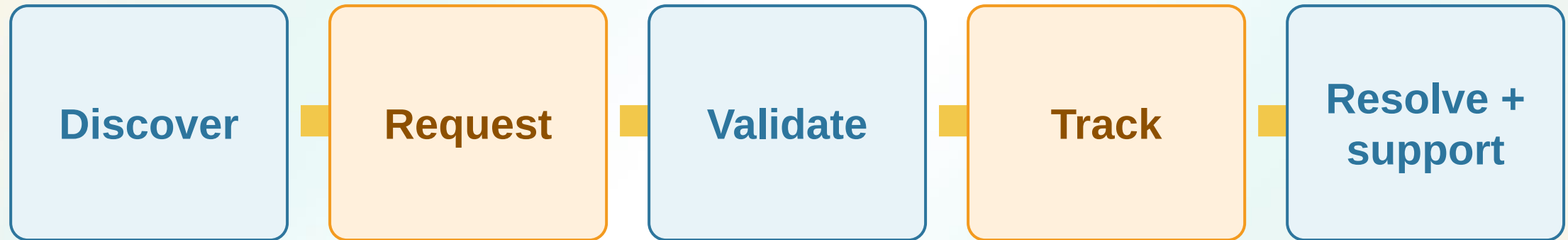


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Web and Mobile lens: follow the service journey

A polished interface cannot repair a service that has unclear ownership, missing feedback, or broken handoffs.



Check accessibility, response time, device constraints, privacy, and recovery from interrupted use.

AGD lens: connect production and player experience

Treat both the production pipeline and the player system as analyzable systems with feedback.

PRODUCTION SYSTEM

**Concept → assets → integration →
testing → release**

**Look for rework, bottlenecks,
version issues, and quality gates.**

PLAYER SYSTEM

**Goal → action → feedback →
challenge → progression**

**Look for usability, balance,
engagement, accessibility, and
learning outcomes.**



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Business Analytics lens: trace data to a decision

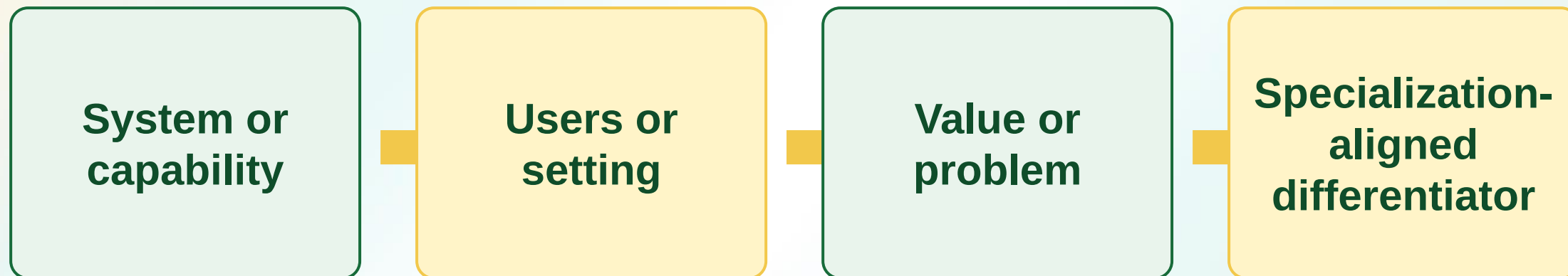
A dashboard is valuable only when data quality, definitions, decision rights, and actions are clear.



Verify ownership, timeliness, bias, privacy, interpretation, and how the insight changes a decision.

Preliminary titles need four essential elements

Write the title after problem framing—not before gathering evidence.



Avoid: feature inventories, unsupported AI claims, vague users, and titles that hide the real transaction.

Title examples: Cybersecurity and Web & Mobile

Cybersecurity: BantayNet: A Secure Incident Reporting and Evidence-Validation System for Public-School Networks

Web and Mobile: LakbayLink: A Mobile Visitor-Service and Local-Guide Coordination Platform for Community Tourism Offices

Check the logic: Each title names the capability, setting, value/problem direction, and specialization lens without becoming a feature list.



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Title examples: AGD and Business Analytics

Animation and Game Development: Storm Shield: A Cooperative Disaster-Preparedness Game for Senior High School Learners

Business Analytics: InsightFlow: A Decision-Support Dashboard for Enrollment and Student-Service Planning

Check the logic: A credible title still requires stakeholder, context, evidence, root-cause, and expected-value support in the portfolio.



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Peer quality gate

Can another team trace your preliminary title back to credible stakeholders, a defined system context, verified root causes, expected value, and a justified specialization lens?



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Complete the Module 1 activity path

Use the Canvas instructions and official templates for the quiz and technical activity.

FORMATIVE ASSESSMENT 1 [AI-PROHIBITED • UNGRADED]

**Complete the Canvas quiz
independently after Module 1.**

**Use the result as evidence of
current mastery.**

CLASS ACTIVITY 1 [AI-INTEGRATED]

**Submit the AI methodology
comparison, verification, critique,
corrected conclusions, validation
record, and AI Use Disclosure.**

Module 1 synthesis

Understand the system: purpose, parts, boundaries, environment, feedback, stakeholders, and organizational context.

Analyze before designing: separate symptoms from root causes and connect evidence to expected value.

Choose responsibly: compare and tailor development approaches; integrate ethics, security, privacy, accessibility, and responsible AI use.

Move the proposal forward: carry the revised context, title logic, methodology rationale, and audit evidence into Module 2.



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REFERENCES

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- **Tilley, S. (2025). Systems analysis and design (13th ed.). Cengage.**
- **Valacich, J. S., & George, J. F. (2024). Modern systems analysis and design (10th ed.). Pearson.**
- **National Institute of Standards and Technology. (2022). Secure Software Development Framework (SSDF), Version 1.1 (NIST SP 800-218).**
- **World Wide Web Consortium. (2023). Web Content Accessibility Guidelines (WCAG) 2.2.**

CARRY YOUR REVISED PROJECT CONTEXT INTO MODULE 2



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